Problem Set 5

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```r
seed_string <- "Blair, Thomas, Aaron, John"
seed_number <-
  seed_string |>
  rlang::hash() |> # hash the string to a hex number
  substr(1, 8) |> # make the number smaller
  strtoi(base = 16) # convert to decimal

set.seed(seed_number)
```

Hamilton's Problems

Exercise 3.1

Exercise 3.2

We can rewrite Y_t in the form

$$\xi_t = \mathbf{F}\xi_{t-1} + \eta_t.$$

Then,

$$\begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix} = \begin{bmatrix} 1.1 & -0.18 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \end{bmatrix} + \begin{bmatrix} \varepsilon_t \\ 0 \end{bmatrix}$$

We find the eigenvalues of $\mathbf{F}$ by solving

$$\begin{aligned} \det \begin{pmatrix} 1.1-\lambda & -0.18 \\ 1 & -\lambda \end{pmatrix} &= 0 \\ -1.1\lambda + \lambda^2 + 0.18 &= 0 \\ \lambda^2 - 1.1\lambda + 0.18 &= 0 \\ (\lambda - 0.9)(\lambda - 0.2) &= 0. \end{aligned}$$

Therefore, the two eigenvalues are $\lambda_1 = 0.2$ and $\lambda_2 = 0.9$. Because

$$max_i|\lambda_i| < 1$$

we know that this AR(2) process admits a covariance-stationary MA(∞) process.

To calculate the mean and autocovariance, we proceed with the eigendecomposition of $\mathbf{F}$, which is:

$$\mathbf{T\Lambda T}^{-1} = \begin{pmatrix} 0.2 & 0.9 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0.2 & 0 \\ 0 & 0.9 \end{pmatrix} \begin{pmatrix} -\frac{10}{7} & \frac{9}{7} \\ \frac{10}{7} & -\frac{2}{7} \end{pmatrix}$$

Then,

$$\begin{aligned} F_{1,1}^j &= 0.2^2 \left(-\frac{10}{7}\right) + 0.9^2 \left(\frac{10}{7}\right) \\ &= \left(\frac{-2}{7}\right) 0.2^j + \left(\frac{9}{7}\right) 0.9^j. \end{aligned}$$

*Authors listed alphabetically by and contributed equally to every question.

If $\psi_j = \left(-\frac{2}{7}\right) 0.2^j + \left(\frac{9}{7}\right) 0.9^j$, then

$$Y_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}.$$

We can see that

$$\mathbb{E}[Y_t] = 0.$$

For the autocovariance,

$$\begin{aligned}
\gamma_j &= \mathbb{E}[(Y_t - \mu)(Y_{t-k} - \mu) \\
&= \mathbb{E}(Y_t Y_{t-k}) = \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}\right)\left(\sum_{l=0}^{\infty} \psi_l \varepsilon_{t-l}\right)\right] \\
&= \mathbb{E}\left[\sum_{n=0}^{\infty}\sum_{j=0}^{n} \left(\psi_j \varepsilon_{t-j}\right)\left(\psi_{n-j}\varepsilon_{t-(n-j)}\right)\right] \text{(by Cauchy product)} \\
&= \sum_{n=0}^{\infty}\sum_{j=0}^{n} \psi_j \psi_{n-j} \mathbb{E}[\varepsilon_{t-j}\varepsilon_{t-n+j}]
\end{aligned}$$

We know that $\mathbb{E}[\varepsilon_{t-j}\varepsilon_{t-n+j}] = 1$ when the indexes are equal to each other $t - j = t - n + j$, which occurs when $n = 2j$. Suppose $n = 0$, then $j = 0$ and we have $\psi_0\psi_0$. Suppose $n = 2$, then $j = 1$ and we have $\psi_1\psi_1$, etc.

Thus, we can observe that the above expression simplifies to

$$\begin{aligned}
\gamma_j &= \sum_{j=0}^{\infty} \psi_j^2 \mathbf{1}\{j \text{ is even}\} \\
&= \sum_{j=0}^{\infty} \left[\left(-\frac{2}{7}\right) 0.2^j + \left(\frac{9}{7}\right) 0.9^j\right]^2 \mathbf{1}\{j \text{ is even}\}
\end{aligned}$$

Excercise 3.3

Excercise 11.5

The Permanent Income Model

1. Finding the Autocovariances

Conditional on the information available at time t, all of the future random variables are mean zero, so the agent's expected income in any period $j \geq 0$ will their current permanent component of income, x_t. The transitory shock includes an ϵ_t that moves current period income from its long run average. So $E_t[y_{t+j+1}] = x_t$. The time $t+1$ expectations are themselves a random variable from the perspective of period t because they depend on the news revealed in period $t+1$. In particular, in period $t+1$, a new value x_{t+1} will be revealed, recentering the agents expectations of all future periods. If $j = 0$, then from period $t+1$'s perspective, y_{t+1} is observed with certainty. But all future $j \geq 1$ are centered at x_{t+1}.

Noting that $x_{t+1} = x_t + \eta_{t+1}$ and $y_{t+1} = x_t + \eta_{t+1} + \varepsilon_{t+1}$, we see that:

$$E_{t+1}[y_{t+j+1}] - E_t[y_{t+j+1}] = \begin{cases} x_t + \eta_{t+1} + \varepsilon_{t+1} - x_t & j = 0 \\ x_t + \eta_{t+1} - x_t & j \geq 1 \end{cases}$$

Which obviously simplifies to:

$$E_{t+1}[y_{t+j+1}] - E_t[y_{t+j+1}] = \begin{cases} \eta_{t+1} + \varepsilon_{t+1} & j = 0 \\ \eta_{t+1} & j \geq 1 \end{cases}$$

With that in mind, we can conveniently express $\Delta c_{t+1} \equiv c_{t+1} - c_t$ as:

$$\begin{aligned}
\Delta c_{t+1} &= (1-\beta) \sum_{j=0}^{\infty} \beta^j \left(E_{t+1}[y_{t+j+1}] - E_t[y_{t+j+1}] \right) \\
&= (1-\beta)\beta^0 \left(E_{t+1}[y_{t+0+1}] - E_t[y_{t+0+1}] \right) + (1-\beta) \sum_{j=1}^{\infty} \beta^j \left(E_{t+1}[y_{t+j+1}] - E_t[y_{t+j+1}] \right) \\
&= (1-\beta) \left(\eta_{t+1} + \varepsilon_{t+1} \right) + (1-\beta) \sum_{j=1}^{\infty} \beta^j \eta_{t+1} \\
&= (1-\beta) \left(\varepsilon_{t+1} \right) + (1-\beta)\eta_{t+1} + (1-\beta) \sum_{j=1}^{\infty} \beta^j \eta_{t+1} \\
&= (1-\beta) \left(\varepsilon_{t+1} \right) + (1-\beta) \sum_{j=0}^{\infty} \beta^j \eta_{t+1} \\
&= (1-\beta) \left(\varepsilon_{t+1} \right) + \eta_{t+1}(1-\beta) \sum_{j=0}^{\infty} \beta^j \\
&= (1-\beta) \left(\varepsilon_{t+1} \right) + \eta_{t+1}(1-\beta) \frac{1}{1-\beta} \\
&= (1-\beta)\varepsilon_{t+1} + \eta_{t+1}
\end{aligned}$$

If we define $u_t = (1-\beta)\varepsilon_{t+1} + \eta_{t+1} \sim \mathcal{N}(0, (1-\beta)^2\sigma_\varepsilon^2 + \sigma_\eta^2)$, then we can think of Δc_t as white noise process with mean 0 and variance: $(1-\beta)^2\sigma_\varepsilon^2 + \sigma_\eta^2)$.

From this, it is clear to see that the IRFs and autocovariances for all $j \neq 0$ will be zero. There is absolutely no information passed between periods (at least for the per-period change in consumption).

If $j = 0$, then we have already expressed the variance as $(1-\beta)^2\sigma_\varepsilon^2 + \sigma_\eta^2)$ Clearly, these are increasing in the variances of the primitives. But interesting as the agent becomes more patient ($\beta \to 1$), the variance in consumption reduces to only variance in the permanent component (because people no longer respond to the transitory shocks.)

And the IRFs are simply the partials with respect to ε which is $(1-\beta)$ and for η which is 1.

Assuming $\beta \in (0, 1)$ (people strictly discount the future), then the magnitude of a transitory shock on the change in consumption is less than the same sized shock to permanent income. The permanent IRFs are invariant to model parameters, whereas as individuals become more patient $\beta \to 1$, the IRF for the transitory shock approaches zero (little change to previous period consumption).

2. The Vector Process

Noting that

$$\begin{aligned}
\Delta y_{t+1} &= (x_{t+1} + \varepsilon_{t+1}) - (x_t + \varepsilon_t) \\
&= (x_t + \eta_{t+1} + \varepsilon_{t+1}) - (x_t + \varepsilon_t) \\
&= \eta_{t+1} + \varepsilon_{t+1} - \varepsilon_t
\end{aligned}$$

Decrementing the indices by one period, we can see that:

$$\begin{bmatrix}\Delta c_t \\ \Delta y_t\end{bmatrix} = \begin{bmatrix}(1-\beta)\varepsilon_t + \eta_t \\ \eta_t + \varepsilon_t - \varepsilon_{t-1}\end{bmatrix}$$

From this form, it would be natural to represent using matrix notation:

$$\mathbf{\Delta}_t = \mathbf{\Psi}_0 \mathbf{u}_t + \mathbf{\Psi}_1 \mathbf{u}_{t-1}$$

Where

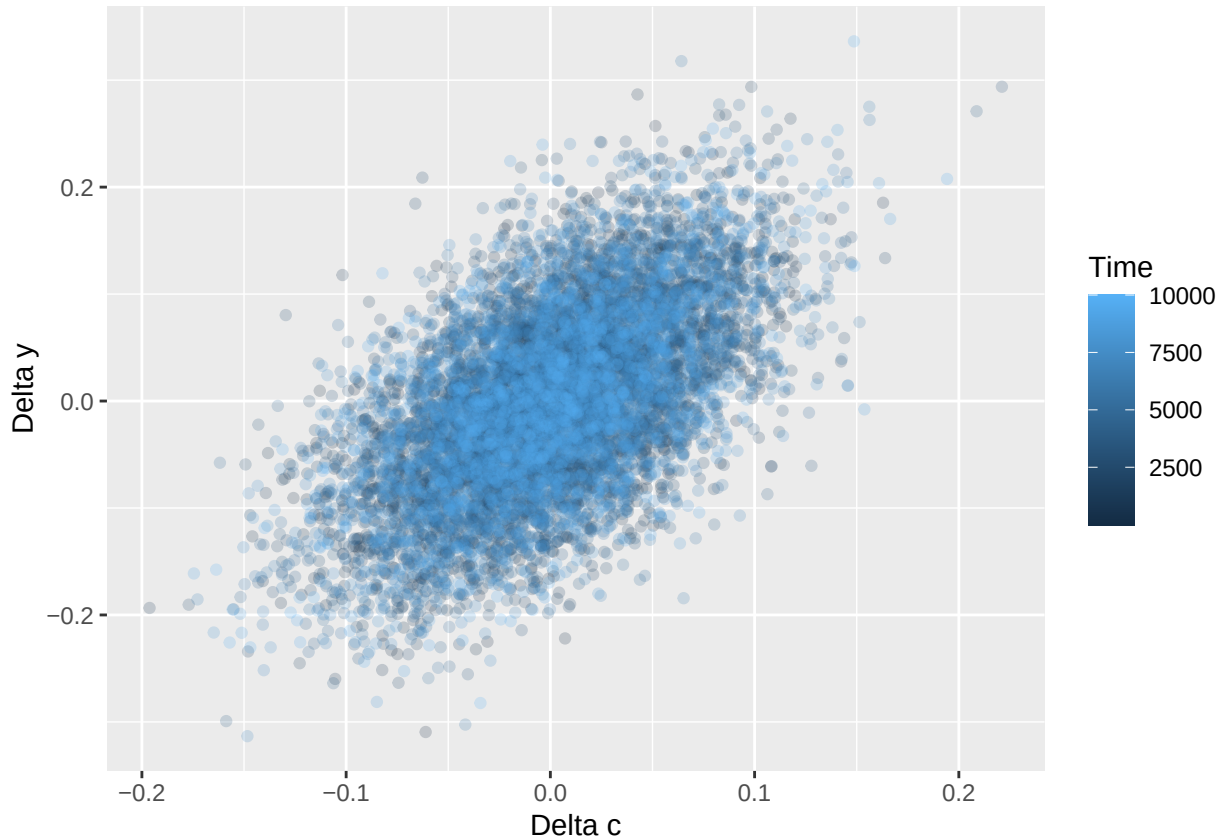
$$\mathbf{\Psi}_0 = \begin{bmatrix}(1-\beta)\sigma_\varepsilon & \sigma_\eta \\ \sigma_\varepsilon & \sigma_\eta\end{bmatrix}, \mathbf{\Psi}_1 = \begin{bmatrix}0 & 0 \\ 0 & -\sigma_\varepsilon\end{bmatrix}$$

and $\mathbf{u}_t$ is a vector of independent white noise with unit variance.

Which is a rather simple vector moving average equation.

3. Simulation Exercise

```
capT = 10e3
beta = .99
sigma_e = .05
sigma_h = .05

e_draws <- rnorm(capT)*sigma_e
h_draws <- rnorm(capT)*sigma_h

rand_mat <- cbind(e = e_draws, h = h_draws)
v_mat <- matrix(nrow = nrow(rand_mat), ncol = 2)
colnames(v_mat) <- c("Delta c","Delta y")

for (i in 1:capT){
  if (i == 1){
    rand_last <- c(e = 0, h = 0)
  }else{
    rand_last <- rand_mat[i-1,]
  }

  rand_cur <- rand_mat[i,]

  v_mat[i, 'Delta c'] <- (1-beta)*rand_cur['e']+rand_cur['h']
  v_mat[i, 'Delta y'] <- rand_cur['h'] + rand_cur['e'] - rand_last['e']
}

library(ggplot2)
ggplot()+
  geom_point(aes(x=v_mat[,1], y = v_mat[,2], col = 1:capT), alpha = .2)+
  labs(color = "Time",
       x = "Delta c",
       y = "Delta y")
```




```r
library(MTS)
m2=VARMA(v_mat,p=0,q=1,include.mean=FALSE)
```

```
## Number of parameters:  4
## initial estimates:  0.0021 0.0038 0.9284 -0.9254
## Par. lower-bounds:  -0.026 -0.0157 0.888 -0.9535
## Par. upper-bounds:  0.0302 0.0234 0.9688 -0.8973
## Final   Estimates:  -0.01779682 0.02340764 0.9604215 -0.9535293
##
## Coefficient(s):
##       Estimate  Std. Error  t value Pr(>|t|)
## [1,]  -0.01780     0.01426   -1.248   0.2121
## [2,]   0.02341     0.01009    2.319   0.0204 *
## [3,]   0.96042     0.01515   63.378   <2e-16 ***
## [4,]  -0.95353     0.01121  -85.023   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## ---
## Estimates in matrix form:
## MA coefficient matrix
## MA( 1 )-matrix
##         [,1]    [,2]
## [1,]  0.0178 -0.0234
## [2,] -0.9604  0.9535
##
## Residuals cov-matrix:
```

```
##             [,1]        [,2]
## [1,] 0.002514670 0.002540158
## [2,] 0.002540158 0.005031062
## ----
## aic=  -11.99031
## bic=  -11.98743
```

```
m2$coef
```

```
##             [,1]       [,2]
## [1,] -0.01779682  0.9604215
## [2,]  0.02340764 -0.9535293
```

```
#matrix(c(1-beta, ))
```

The Kalman Filter

1. Analytical Moments

We are given the ARMA(1,1) process:

$$y_t = \phi y_{t-1} + \varepsilon_t + \gamma \varepsilon_{t-1}$$

In lag operator notation, we have:

$$\begin{aligned} y_t &= \phi L y_t + \varepsilon_t + \gamma L \varepsilon_t \\ (1-\phi L) y_t &= (1+\gamma L)\varepsilon_t \end{aligned}$$

The multiplicative inverse of $(1-\phi L)$ is $(1+\phi L+\phi^2 L^2 - \ldots)$, and so we can alternately express the ARMA process as an MA(∞):

$$\begin{aligned} (1+\phi L+\phi^2 L^2 - \ldots)(1-\phi L) y_t &= (1+\phi L+\phi^2 L^2 - \ldots)(1+\gamma L)\varepsilon_t \\ y_t &= (1+\phi L+\phi^2 L^2 - \ldots)(1+\gamma L)\varepsilon_t \\ &= [(1+\phi L+\phi^2 L^2 - \ldots) + (\gamma L + \gamma\phi L + \gamma\phi^2 L^2 + \ldots)]\varepsilon_t \\ &= [\underbrace{1}_{\psi_0} + \underbrace{\phi^0(\phi+\gamma)}_{\psi_1} L - \underbrace{\phi^1(\phi+\gamma)}_{\psi_2} L^2 - \underbrace{\phi^2(\phi+\gamma)}_{\psi_3} L^3 + \ldots]\varepsilon_t \\ &= \psi(L)\varepsilon_t \end{aligned}$$

Assuming at every point that each step is well defined.

In this form, it is much easier to compute the various moments of the distribution.

Namely we have:

$$\begin{aligned} E[y_t] &= E\left[[1+\phi^0(\phi+\gamma)L + \phi^1(\phi+\gamma)L^2 + \phi^2(\phi+\gamma)L^3 - \ldots]\varepsilon_t\right] \\ &= [1+\phi^0(\phi+\gamma)L + \phi^1(\phi+\gamma)L^2 + \phi^2(\phi+\gamma)L^3 - \ldots]E[\varepsilon_t] \\ &= [1+\phi^0(\phi+\gamma)L + \phi^1(\phi+\gamma)L^2 + \phi^2(\phi+\gamma)L^3 - \ldots]0 \\ &= 0 \end{aligned}$$

The variance of any $MA(\infty)$ process is given by:

$$
\begin{aligned}
\gamma_0 &= \sigma^2 \sum_{j=0}^{\infty} \psi_j^2 \\
&= \sigma^2 \left(\psi_0^2 + \sum_{j=1}^{\infty} \psi_j^2 \right) \\
&= \sigma^2 \left(1^2 + \sum_{j=1}^{\infty} (\phi^{j-1}(\phi+\gamma))^2 \right) \\
&= \sigma^2 \left(1 + (\phi+\gamma))^2 \sum_{j=1}^{\infty} \phi^{j-1\,2} \right) \\
&= \sigma^2 \left(1 + (\phi+\gamma))^2 \sum_{j=0}^{\infty} \phi^{j\,2} \right) \\
&= \sigma^2 \left(1 + (\phi+\gamma))^2 \sum_{j=0}^{\infty} \phi^{2j} \right) \\
&= \boxed{\sigma^2 \left(1 + (\phi+\gamma))^2 \frac{1}{1-\phi^2} \right)}
\end{aligned}
$$

requiring that $\phi^2 < 1 \implies |\phi| < 1$

The jth autocovariance of the MA(∞) is given by:

$$
\begin{aligned}
\gamma_j &= \sigma^2 \sum_{k=0}^{\infty} \psi_k \psi_{j+k} \\
&= \sigma^2 \left(\psi_0 \psi_j + \sum_{k=1}^{\infty} \psi_k \psi_{j+k} \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + \sum_{k=1}^{\infty} \phi^{k-1}(\phi+\gamma)\phi^{j+k-1}(\phi+\gamma) \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + \sum_{\kappa=0}^{\infty} \phi^{\kappa}(\phi+\gamma)\phi^{j+\kappa}(\phi+\gamma) \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \sum_{\kappa=0}^{\infty} \phi^{\kappa}\phi^{j+\kappa} \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \sum_{\kappa=0}^{\infty} \phi^{2\kappa}\phi^{j} \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \phi^{j} \sum_{\kappa=0}^{\infty} \phi^{2\kappa} \right) \\
&= \sigma^2 \left(1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \phi^{j} \frac{1}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left(1 + (\phi+\gamma)\phi \frac{1}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left(1 + (\phi+\gamma) \frac{\phi}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left(\frac{1-\phi^2}{1-\phi^2} + \frac{\phi^2+\phi\gamma}{1-\phi^2} \right) \\
&= \boxed{\sigma^2 \phi^{j-1}(\phi+\gamma) \left(\frac{1+\phi\gamma}{1-\phi^2} \right)}
\end{aligned}
$$

Truly a bizarre expression that leaves little room for intuition. However, the series convergence there also only required that $\phi^2 < 1$, so $|\phi| < 1$ is still sufficient to ensure covariance stationarity.

2. Conditional Expectations

3. Conditional Likelihood

4. State Space Representation

A state space representation is a system of vector equations of the form:

$$
\begin{aligned}
y_t &= \mu + A z_t + H \eta_t \\
z_t &= \Phi z_{t-1} + R \xi_t
\end{aligned}
$$

Where $\eta_t \sim \mathcal{N}(0, I), \xi_t \sim \mathcal{N}(0, \Sigma)$. Letting $z_t = \begin{bmatrix} y_t & \varepsilon_t \end{bmatrix}'$ we can represent the observation equation as:

$$
y_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_t \\ \varepsilon_t \end{bmatrix} + 0\eta_t
$$

The η component will not be needed, so we can set $H = 0$. Here's a canonical way to represent the state equation that ensures dimensional conformability at every step.

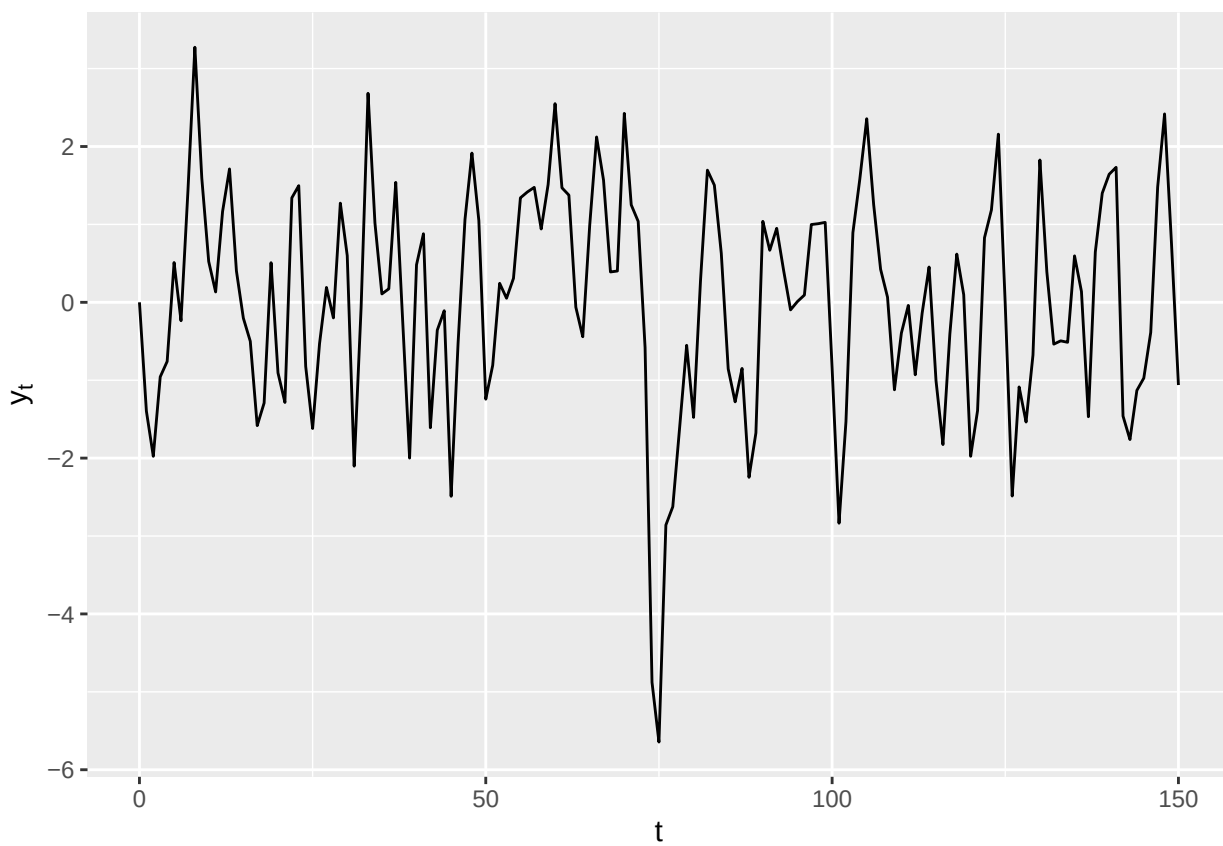
$$\begin{bmatrix} y_t \\ \varepsilon_t \end{bmatrix} = \begin{bmatrix} \phi & \gamma \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ \varepsilon_{t-1} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \xi_t$$

Where ξ_t is a 2-long vector distributed $\xi_t \sim \mathcal{N}(0, \sigma^2 I)$. You could also use a one dimensional ξ_t and then just the first column of the matrix before ξ_t.

5. Simulation

```
capT = 150
# structural parameters
phi = .9
gamma = .1
sigma_squared = 1

# state space parameters
dim_state <- 2
Sigma = sigma_squared * diag(nrow = dim_state)
R = matrix(c(1, 0,
             1, 0),
           byrow = T, nrow = dim_state)
Phi = matrix(c(gamma, phi,
               0, 0),
             byrow = T, nrow = dim_state)

# Observation parameters
dim_obs <- 1 #
mu <- c(0)
A <- matrix(c(1, 0), byrow = T, nrow = dim_obs, ncol = dim_state)
H <- matrix(c(0), byrow = T, nrow = dim_obs)

evolve_state <- function(z, Phi, R, Sigma){
  # compute multivariate normal using linearity
  xi = Sigma%*%rnorm(nrow(Sigma))

  (Phi %*% z) + R%*%xi
}

evolve_observations <- function(z, mu, A, H){
  eta <- rnorm(nrow(H))

  mu + (A %*% z) + H%*%eta
}

z0 = c(0,0)
simulate <- function(capT, z0, Phi, R, Sigma, mu, A, H){
  # loop capT times, feeding the result of the last iteration into
  # this this iteration
  states <- purrr::accumulate(
  1:capT,
  ~ evolve_state(.x, Phi, R, Sigma),
  .init = z0
)
```

```r
  # compute observation from latent state
  observations <- lapply(states, function(z){
    evolve_observations(z, mu, A, H)
  })

  # return long matrices with one row per time step
  states <- do.call(cbind, states) |> t()
  observations = do.call(cbind, observations) |> t()


  return(list(states = states, observations = observations))
}

sim_results <- simulate(capT, c(0,0), Phi, R, Sigma, mu, A, H)

library(ggplot2)
ggplot()+
  geom_line(aes(x=0:(nrow(sim_results$observations)-1),
                y=sim_results$observations[,1]))+
  labs(y = expression("y"[t]*""),
       x = "t")
```



6. Implementing the Kalman Filter

```r
predict_y <- function(z, mu, A){
  mu + (A %*% z)
}

compute_v <- function(y, z, mu, A){
 y - predict_y(z, mu, A)
}

v_posterior_variance <- function(P, A, H){
  (A %*% P %*% t(A)) + H%*%t(H)
}

prediction_error_variance <- function(z, P, A){
  E
}
```

7. Applying the Kalman Filter

Metropolis Hastings

1. Numerical Maximization

2. Generating M-H Draws

3. Lots of Simulations